Madeleine S. Yuh

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EDUCATION:

Ph.D., Mechanical Engineering, Purdue University – West Lafayette **May 2026**
Area: Systems, Measurements, and Control, **Advisor**: Dr. Neera Jain

M.S., Mechanical Engineering, Purdue University – West Lafayette **Aug 2026**
Area: Systems, Measurements, and Control, **Advisor**: Dr. Neera Jain

B.S., Mechanical Engineering, University of Illinois at Chicago **May 2020**
Summa Cum Laude, UIC Honors College

RESEARCH INTERESTS:

Deterministic & Probabilistic Dynamic Modeling, Optimal Control, Human-Machine Interaction, Machine Learning, Reinforcement Learning, Cyber-Physical Systems, Intelligent Tutoring Systems, Human Automation Teaming, Rehabilitation Robotics.

RELEVANT COURSEWORK:

Multidisciplinary Design Optimization, Optimal Control & Estimation, Nonlinear Feedback Control, Digital Signal Processing, Machine Learning, Reinforcement Learning, System Identification, Statistics.

HONORS & AWARDS:

1. Frederick N. Andrews Fellowship, School of Mechanical Engineering at Purdue University, 2020–2024
2. 2022 CPHS Fellows, 4th IFAC Workshop on Cyber-Physical Human Systems, 2022
3. ASPIRE 2024 Alphonse Chapanis Best Student Paper Award, 2024
4. HFES CEDM Technical Group Best Student Paper Award Finalist, 2024
5. Hommert Engineering Excellence Fellowship, 2024
6. Ward A. Lambert Graduate Teaching Fellowship, 2024
7. ICON Student Research Conference 2025 Best Presentation Award, 2025
8. 2025 NSF CPS Rising Star, 2025
9. Purdue OGSPS Mentoring Award for Graduate Students College of Engineering nominee, 2025
10. MECC 2025 DSCD Rising Star, 2025
11. MIT Rising Stars in Mechanical Engineering, 2025
12. Summer Undergraduate Research Fellowship graduate mentor award, Purdue University, 2022 & 2025

PUBLICATIONS:

Peer Reviewed Journal Articles

- K. J. Williams, **M. S. Yuh**, and N. Jain, "A Computational Model of Coupled Human Trust and Self-Confidence Dynamics", *ACM Transactions on Human-Robot Interaction,* vol. 12, no. 3, pp.1-29, June 2023.
- **M. S. Yuh**, K. R. Ortiz, K. Sommer-Kohrt, M. Oishi and N. Jain, "Classification of Human Learning Stages via Kernel Distribution Embeddings," *IEEE Open Journal of Control Systems*, pp. 1-16, January 2024.
- **M. S. Yuh** and N. Jain, "A Personalized and Adaptive Intelligent Tutoring System for Psychomotor Learning," 2026. [Online]. Available: http://arxiv.org/abs/2610.33467 ***(In Progress)***

Peer Reviewed Conference Papers

- **M. S. Yuh**, S. Byeon, I. Hwang, and N. Jain, "A Heuristic Strategy for Cognitive State-based Feedback Control to Accelerate Human Learning," in *IFAC-PapersOnLine*, vol. 55, no. 41, pp. 107-112, 2022.
- **M. S. Yuh**, E. Rabb, A. Thorpe, and N. Jain, "Using Reward Shaping to Train Cognitive-based Control Policies for Intelligent Tutoring Systems", in *2024 American Control Conference (ACC*), Toronto, ON, 2024.
- **M. S. Yuh** and N. Jain, "Online Self-Confidence Calibration for Improving Learning Outcomes via Intelligent Tutoring Systems," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, p. 10711813241260743, Aug. 2024. *(ASPIRE 2024 Alphonse Chapanis Best Student Paper Award, HFES Cognitive Engineering and Decision Making Technical Group Best Student Paper Award Finalist)*

- **M. S. Yuh** and N. Jain, "Designing Cognitively-Aware Psychomotor Intelligent Tutoring Systems via Multi-Objective and Human-Centric Optimization" in *2025 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Eindhoven, the Netherlands, 2025.
- M. Wielatz, M. Pandya, **M. S. Yuh**, and N. Jain, "The Role of Dispositional Trust in Adaptive Automation for Trust Calibration" in *2025 IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, Eindhoven, the Netherlands, 2025.
- Y. Shao, **M. S. Yuh,** and N. Jain, "Learning What Learners Want: Using Inverse Reinforcement Learning for Tutoring Policy Design in Intelligent Tutoring Systems" in *2026 IFAC Workshop on Cyber-Physical human Systems (CPHS 2026)*, Redondo Beach, California, USA, 2026*.* ***(Accepted)***

Conference Papers

- K. Ortiz, J. Hunter, A. Thorpe, **M. S. Yuh**, T. Reid, N. Jain and M. Oishi, "Assessing the Relationship Between Learning Stages and Prefrontal Cortex Activation in a Psychomotor Task," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, p. 10711813241260675, Aug. 2024
- S. Byeon, **M. S. Yuh**, J. Choi, N. Jain, and I. Hwang, "Workload Classification for Function Allocations in Human-Autonomy Teaming Using Noninvasive Measurements," in AIAA SCITECH 2025 Forum, Jan. 2025.

Book Chapters

- N. Jain, T. Reid, K. Akash, **M. S. Yuh**, and J. Hunter, "Enabling Human-Aware Autonomy Through Cognitive Modeling and Feedback Control," in Cyber-Physical-Human Systems: Fundamentals and Applications. A. M. Annaswamy, P. P. Khargonekar, F. Lamnabhi-Lagarrigue, and S. K. Spurgeon, Eds. Hoboken, NJ, USA; John Wiley & Sons, Inc. 2023, ch. 5, pp 95-124.

INVITED TALKS:

- "Designing Cognitively Aware Intelligent Tutoring Systems via Model-Based Cognitive Control," Seminar Series, Air Force Research Laboratories, Kirtland Airforce Base, August 29, 2024.
- "Designing Cognitively-Aware Psychomotor Intelligent Tutoring Systems via Multi-Objective and Human-Centric Optimization," 711th Human Performance Wing, Air Force Research Laboratory, Wright-Patterson Air Force Base, March 6, 2026.

RESEARCH AND TEACHING EXPERIENCE:

Purdue University – West Lafayette, Indiana

Post-Doctoral Researcher **Aug 2026 – Present**

- Developing and implementing cognitive models of human cognitive states to support adaptive Human-Autonomy Teaming (HAT) and improve human–autonomy interaction.
- Investigating optimal control policies for Human-Autonomy Teaming based on auctioning algorithms for dynamic task allocation between humans and autonomous agents.
- Integrating human cognitive state models with autonomy and control frameworks to enable adaptive, human-centered decision-making in Human-Autonomy Teaming.

Instructor **Aug 2026 – Present**

- Teaching two sections of Systems, Measurements, and Controls 1 (ME 36500) in the Fall 2027 semester, providing instruction in fundamental systems and control engineering concepts.

Research Assistant **Aug 2020 – July 2026**

- Formulated and validated probabilistic model of human states of trust and self-confidence as a dynamic system when interacting with an autonomous system.
- Designed and experimentally validated control algorithms for achieving calibration of human cognitive states in psychomotor learning contexts.
- Synthesized and implemented cognitive-based control policies using reward shaping in an intelligent tutoring system for a psychomotor task.
- Specialization in user studies and control algorithms to infer human cognitive states using behavioral and psycho-physiological data.

- Identified and characterized learning stages quantitatively for psychomotor learning task as well as developed an online learning stage classifier to provide learning stages as inputs to future cognitive control policies.

Ward A. Lambert Graduate Teaching Fellowship **Jan 2025 – Dec 2025**
- Instructor for a section of Systems, Measurements, and Controls 1 (ME 36500) in the Fall 2025 Semester
- Shadowed a faculty member during the spring 2025 semester teaching a section of Systems, Measurements, and Controls 1 (ME 36500).

Midwest Orthopaedics at Rush – Chicago, Illinois
Undergraduate Research Assistant **Sep 2017 – Nov 2018**
- Experimentally validated finite element analysis (FEA) simulation data on wear that occurs due to forces experienced in the knee throughout life in metal/poly-ethylene knee implants. Worked with wear testing machines called pin-on-disc and wheel-on-flat to simulate movements of the knee over 1 million cycles.
- Simulated stresses in patellofemoral joint dependent on patella tendon on tibial tubercle position using FEA.
- Used Geomagic Wrap and Truegrid software to smooth and mesh CT scans of bones, cartilage, and tendons.
- Used Abaqus to assemble parts of the Patellofemoral joint to make a model for simulation and FEA.

INTERNSHIPS:

Sandia National Laboratories – Albuquerque, New Mexico
AI for Autonomy Intern **June 2024 – Feb 2026**
- Designed, simulated, and implemented auctioning algorithms for task-allocation in multi-agent swarm settings.
- Performed system identification to validate laboratory experiments.

PROFESSIONAL ACTIVITIES & SERVICE:

Purdue ICON Student Research Conference Program Coordinator **Nov 2022 – Apr 2024**
- **Program Coordinator:** Project lead on coordinating student research conference committee members with roles and reviewing submitted abstracts for presentation and posters. Created website with relevant information pertaining to scheduling, registration, and abstract submission.
- **Poster Session Cochair:** Organized poster session for the 2023 and 2024 Purdue ICON student research conference for students to offer students the chance to present their research and network with faculty members and keynote speakers and explore new research avenues and interdisciplinary partnerships.

Purdue ME Taskforce on Female Undergraduate Recruitment **Aug 2021 – Jan 2024**
- Graduate student representative of ME Taskforce on Female Undergraduate Recruitment.
- Served as panelist for Women in Mechanical Engineering Symposium for current women in ME student panel.
- Organized poster session for the Women in Mechanical Engineering Symposium to encourage Purdue undergraduates to explore different research topics and consider graduate school after graduation.

SIRI 2023 Graduate Leadership Team **May 2023- Aug 2023**
- Organized 8-week undergraduate research program curriculum to facilitate meaningful research experience aimed towards under-represented students interested in cyber-physical systems.

MENTORSHIP:

Undergraduate Student Mentor, Purdue University **Aug 2022 – Present**
- Mentored student working on incorporated feedback in an online drone simulation generated through GPT-4 to perturb human self-confidence while learning. This data will be used to extend an existing probabilistic model of human self-confidence behavior and control policy synthesis.
- Mentored student working on further developing an online drone simulation to capture human risk behavior.
- Mentored student running user study on further validating heuristic strategy used to decide when to provide autonomous assistance to users while calibrating self-confidence to skill in learning context. Eye-tracking data used to analyze factors that impact self-confidence and identify learning stages in a drone simulator.

SURF/SIRI Undergraduate Research Student Mentor, Purdue University **May 2021 – Aug 2025**

- **SURF Summer 2025:** Mentoring two students to incorporate haptic feedback into a psychomotor intelligent tutoring system to improve student-tutoring agent communication.
- **SURF Summer 2024:** Mentored student who worked on online updating of probabilistic cognitive state models of human learners as they interact with an intelligent tutoring system for a psychomotor learning task.
- **SURF Summer 2023:** Mentored student on developing an experimental platform and probabilistic model that captures self-confidence and workload perturbations in human-automation teaming contexts.
- **SURF Summer 2022:** Mentored student who incorporated and analyzed eye-tracking data in an in-person user study where users learn to fly a drone in a simulator. Data was used to infer human self-confidence cognitive states.
- **SIRI Summer 2022:** Mentored students on developing a probabilistic model on the dynamics of human self-confidence cognitive states when interacting with autonomous assistance in drone landing simulator module.
- **SIRI Summer 2021:** Mentored students on designing an online based drone simulation experimental platform designed to collect human cognitive and behavioral data when interaction with automation through autonomous assistance.